Daksh Kohli

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Education

Bachelor of Engineering in Computer Engineering *2024 – 2028*
Thapar Institute of Engineering and Technology, Patiala CGPA: 8.39

Leadership & Experience

Technical Intern – TSLAS Office, TIET *2026 – Present*

- Co-developed an auto-scheduling engine as part of a 2-person team, generating timetables across 300+ major, minor, and elective combinations for institute-wide faculty use, under the Dean, TSLAS, using Python and simulated annealing-based optimization.
- Streamlined manual scheduling workflows across multiple departments, cutting redundant operational effort and freeing up team bandwidth for planning and execution initiatives.

Technical Secretary – Microsoft Learn Student Chapter (MLSC), TIET *2025 – Present*

- Lead technical initiatives and cross-functional collaboration across a 14-member executive board and 40+ contributors, driving execution on 5+ concurrent projects.
- Mentored 30+ students across 3 hackathons, and manage task delegation, sprint planning, and team communication across 4+ concurrent technical projects.

Projects

Constraint-Based Timetable Scheduling System – TSLAS Office, TIET
Python, Next.js, AWS, DynamoDB, Docker

- Engineered a production-grade timetable scheduling system for 800+ students and 30+ faculty members, deployed on AWS with DynamoDB, supporting 20+ configurable academic constraints.
- Implemented a simulated annealing optimization algorithm to resolve faculty overlaps, room-capacity limits, elective clashes, and cross-department timetable conflicts.
- Designed and integrated a CI/CD pipeline with 11 automated validation tests covering room, faculty, and student clashes to ensure timetable integrity before deployment.

Thapar Library Management System – Official Institute Project
Tech Lead
Next.js, FastAPI, PostgreSQL, Redis, Docker, Nginx

- Leading a 7-member development team as Tech Lead to rebuild Thapar Institute's official library management platform, targeting 10,000+ monthly active users.
- Architected the system with Redis-backed real-time updates and concurrency control to safely handle simultaneous book checkouts across 10,000+ monthly users, while owning sprint planning and code review as Tech Lead.

DocSearch RAG – Hybrid Retrieval (BM25 + Vector) Document Q&A GitHub Link
Java, Spring Boot, PostgreSQL, pgvector, ONNX Runtime, Ollama, Docker

- Built a RAG system over PDF, Word, PowerPoint, Excel, and CSV files: hand-written tokenizer, Porter stemmer, and positional inverted index for BM25, fused with pgvector semantic search (MiniLM via ONNX Runtime) using Reciprocal Rank Fusion and heuristic reranking.
- Added streamed answers from a local LLM (Ollama) with page/slide-level citations, an evidence gate that skips the LLM when no relevant context is found, and an extractive fallback; built a Recall@k/MRR evaluation harness and 50 unit tests.

Volatility Regime Forecasting with Deep Learning GitHub Link
Python, PyTorch, LSTM/Transformer, Scikit-learn, GARCH, SHAP

- Built a leakage-aware pipeline forecasting next-20-day SPY volatility regimes from 100 features across 9 ETFs, using forward-looking labels, train-only GMM labeling and scaling, and purged walk-forward validation.
- Benchmarked Logistic Regression, GBT, LSTM, and Transformer against Persistence and GJR-GARCH baselines, plus a cost-aware volatility-targeting strategy; added SHAP explainability and 40 tests guarding against leakage.

Technical Skills

Languages: Python, Java, C++, JavaScript
Backend & APIs: FastAPI, Spring Boot, Node.js, Express.js, REST API Design, RESTful Services
Frontend: Next.js, React.js, Tailwind CSS, Framer Motion, Three.js, React Three Fiber
Databases: PostgreSQL, pgvector, MySQL, MongoDB, Redis
DevOps & Infrastructure: AWS, DynamoDB, Docker, Nginx, Cloudflare, Railway, CI/CD, Git, GitHub, Postman
Machine Learning: PyTorch, LSTM, Transformers, Scikit-learn, GMM Clustering, RAG, Hybrid Retrieval, TF-IDF/BM25, Semantic Search
Auth & Platform Services: Clerk
Core Concepts: Database Management Systems (DBMS), Object-Oriented Programming (OOP), System Design, Data Modeling, Transaction Management, Concurrency Control, API Design
Project Management & Collaboration: SCRUM, JIRA, Agile Workflow, Technical Team Leadership, Stakeholder Communication, Cross-Functional Collaboration, Mentorship